9/15/2025

Financial Accounting

Final Assignment

O. P. JINDAL GLOBAL UNIVERSITY

QUESTION 1 ~BIZGROW SOLUTIONS~

1 TRANSACTION EFFECTS (ASSETS = LIABILITIES + EQUITY)

1. **Aug 1**

Alex Chen invested ₹500,000 cash to start the business.

Cash ↑ ₹500,000 → Assets ↑ ₹500,000

Owner's Capital ↑ ₹500,000 → Equity ↑ ₹500,000

2. **Aug 2**

Office furniture purchased on credit for ₹75,000.

Furniture ↑ ₹75,000 → Assets ↑ ₹75,000

Accounts Payable ↑ ₹75,000 → Liabilities ↑ ₹75,000

3. **Aug 5**

Paid office rent of ₹20,000 in cash.

Rent Expense ↑ ₹20,000 → Equity ↓ ₹20,000

Cash ↓ ₹20,000 → Assets ↓ ₹20,000

4. **Aug 10**

Provided consulting services worth ₹120,000. Received ₹70,000 cash, and ₹50,000 remains receivable.

Cash ↑ ₹70,000 → Assets ↑ ₹70,000

Accounts Receivable ↑ ₹50,000 → Assets ↑ ₹50,000

Service Revenue ↑ ₹120,000 → Equity ↑ ₹120,000

5. **Aug 15**

Paid ₹25,000 cash for salaries.

Salaries Expense ↑ ₹25,000 → Equity ↓ ₹25,000

Cash ↓ ₹25,000 → Assets ↓ ₹25,000

6. **Aug 18**

Received ₹30,000 in advance for future consulting services.

Cash ↑ ₹30,000 → Assets ↑ ₹30,000

Unearned Revenue ↑ ₹30,000 → Liabilities ↑ ₹30,000

7. **Aug 22**

Paid ₹40,000 to reduce the furniture payable balance.

Cash ↓ ₹40,000 → Assets ↓ ₹40,000

Accounts Payable ↓ ₹40,000 → Liabilities ↓ ₹40,000

8. **Aug 25**

Collected ₹35,000 cash from receivables.

Cash ↑ ₹35,000 → Assets ↑ ₹35,000

Accounts Receivable ↓ ₹35,000 → Assets ↓ ₹35,000

9. **Aug 28**

Paid ₹15,000 cash for advertising expenses.

Advertising Expense ↑ ₹15,000 → Equity ↓ ₹15,000

Cash ↓ ₹15,000 → Assets ↓ ₹15,000

10. Aug 30

Alex Chen withdrew ₹10,000 for personal use.

Owner's Drawings ↑ ₹10,000 → Equity ↓ ₹10,000

Cash ↓ ₹10,000 → Assets ↓ ₹10,000

2 ENDING BALANCES BY ACCOUNT (AFTER ALL 10)

Assets

• Cash = ₹525,000

• Accounts Receivable = ₹15,000

• Furniture = ₹75,000

Total Assets = ₹615,000

Liabilities

• Accounts Payable = ₹35,000

• Unearned Revenue = ₹30,000

Total Liabilities = ₹65,000

Equity (permanent + contra)

• Owner's Capital = ₹500,000

• Owner's Drawings = ₹10,000 (contra)

Temporary accounts (for the month)

• Service Revenue = ₹120,000

• Expenses: Rent ₹20,000; Salaries ₹25,000; Advertising ₹15,000 → Total Expenses ₹60,000

• Net Income = ₹60,000

Equity reconciliation (to validate)

Owner's Capital ₹500,000 + Net Income ₹60,000 − Drawings ₹10,000 = **Ending Equity** ₹550,000

Check

Assets ₹615,000 = Liabilities ₹65,000 + Ending Equity ₹550,000 → ₹615,000

3 Income Statement — Month ended Aug 31, 2025

Revenue

• Consulting Revenue = ₹120,000

Expenses

• Rent = ₹20,000

• Salaries = ₹25,000

• Advertising = ₹15,000

Total Expenses = ₹60,000

Net Income = ₹60,000

4 Balance Sheet — As of Aug 31, 2025

Assets

• Cash = ₹525,000

• Accounts Receivable = ₹15,000

• Furniture = ₹75,000

Total Assets = ₹615,000

Liabilities

• Accounts Payable = ₹35,000

• Unearned Revenue = ₹30,000

Total Liabilities = ₹65,000

Owner's Equity

• Owner's Capital = ₹500,000

• Add: Net Income = ₹60,000

• Less: Drawings = ₹10,000

Ending Owner's Equity = ₹550,000

Total Liabilities and Equity = ₹615,000

EXECUTIVE SUMMARY

BIZGROW SOLUTIONS

FINANCIAL PERFORMANCE (AUGUST 2025)

- Business startred with ₹500,000 owner investment
- Provided consulting services worth ₹120,000, collected ₹105,000 in cash by month
- Key expenses: Rent: ₹20,000, Salaries ₹25,000, Advertising ₹15,000
- Owner withdrew ₹10,000 for st0C
- Unearned revenue ₹30,000 for September services

NET RESULTS (AUGUST 2025)

- Total Assets ₹475,000
- Liabilities ₹35,000
- Equity ₹440,000
- Net Income ₹60,000

INTERPRETATION

BizGrow Solutions has forn a strong start with positive net income in its first month. The main strenngeus are revenue generaton and equity growth.

QUESTION 2 ~THE GADGET HUB~ — OCTOBER 2025 (PERPETUAL INVENTORY)

Key facts

- Beginning inventory: 10 units @ ₹8,000 = ₹80,000
- Purchases: Oct 5 → 20 @ ₹8,500; Oct 15 → 15 @ ₹9,000; Oct 25 → 10 @ ₹9,200
- Sales: Oct 10 → 15 units; Oct 20 → 25 units
- Selling price (uniform) = ₹12,000 per unit
- Total units available = 10 + 20 + 15 + 10 = 55 units
- Units sold = 15 + 25 = 40 units → Ending units = 15

1 FIFO (PERPETUAL)

Inventory layers (chronological):

- Start: 10 @ ₹8,000
- After Oct 5: +20 @ ₹8,500 → layers: 10@8,000; 20@8,500

Oct 10 — sold 15

- Use 10 from 10@8,000 → 10 × 8,000 = ₹80,000
- Use 5 from 20@8,500 → 5 × 8,500 = ₹42,500
- **COGS (Oct 10)** = ₹80,000 + ₹42,500 = **₹122,500**
- Remaining layers after sale: 15 @ ₹8,500

Oct 15 — purchase 15 @ ₹9,000

- Layers now: 15@8,500; 15@9,000

Oct 20 — sold 25

- Use 15 from 15@8,500 → 15 × 8,500 = ₹127,500

- Use 10 from 15@9,000 → 10 × 9,000 = ₹90,000
- **COGS (Oct 20)** = ₹127,500 + ₹90,000 = **₹217,500**
- Remaining layer after sale: 5 @ ₹9,000

Oct 25 — purchase 10 @ ₹9,200

- Ending layers: 5@9,000; 10@9,200

FIFO totals

- **Total COGS (FIFO)** = ₹122,500 + ₹217,500 = **₹340,000**
- **Ending inventory (FIFO)** = (5 × 9,000) + (10 × 9,200) = ₹45,000 + ₹92,000 = **₹137,000**

Check: Cost of goods available = ₹477,000 → 477,000 = 340,000 + 137,000

2 WEIGHTED-AVERAGE (PERPETUAL)

We recalculate average unit cost each time we purchase and use that current average for subsequent sales. Values shown rounded to two decimals; calculations were kept at full precision.

Start

- 10 @ ₹8,000 → total cost ₹80,000 → avg cost ₹8,000.00

Oct 5 — purchase 20 @ ₹8,500

- New total units = 30; total cost = 80,000 + 170,000 = ₹250,000
- **New Avg cost** = 250,000 / 30 = **₹8,333.33** per unit

Oct 10 — sold 15

- **COGS (Oct 10)** = 15 × 8,333.33 = **₹125,000.00**
- Remaining: 15 units, cost left ₹125,000

Oct 15 — purchase 15 @ ₹9,000

- New total units = 30; total cost = 125,000 + 135,000 = ₹260,000
- **New Avg cost** = 260,000 / 30 = **₹8,666.67** per unit

Oct 20 — sold 25

- **COGS (Oct 20)** = 25 × 8,666.67 = **₹216,666.67**
- Remaining: 5 units, cost left ≈ ₹43,333.33

Oct 25 — purchase 10 @ ₹9,200

- New total units = 15; total cost = 43,333.33 + 92,000 = **₹135,333.33**
- **Ending avg cost** = 135,333.33 / 15 = **₹9,022.22** per unit

Weighted-average totals

- **Total COGS (WA)** = 125,000.00 + 216,666.67 = **₹341,666.67**
- **Ending inventory (WA)** = 15 × 9,022.22 = **₹135,333.33**

Check: Cost of goods available = ₹477,000 → 477,000 = 341,666.67 + 135,333.33

3 PROFITABILITY COMPARISON

Sales revenue = 40 units × ₹12,000 = **₹480,000**

Gross profit (FIFO) = Revenue − COGS(FIFO)
= 480,000 − 340,000 = **₹140,000**

Gross profit (Weighted-average) = Revenue − COGS(WA)
= 480,000 − 341,666.67 = **₹138,333.33**

Which method gives higher profit and why

- FIFO gives the higher gross profit by **₹1,666.67**.
- Reason: purchase costs rose during October (8,000 → 8,500 → 9,000 → 9,200). FIFO charges the oldest, lower costs to COGS first, producing a lower COGS and therefore higher gross profit when prices are rising. Weighted average smooths costs across units, producing slightly higher COGS and a lower gross profit in this rising-cost environment.

Final numbers (rounded to two decimals where needed)

- Units sold: 40; Units ending: 15
- Total cost of goods available: **₹477,000**

FIFO

- COGS = **₹340,000.00**
- Ending inventory = **₹137,000.00**
- Gross profit = **₹140,000.00**

Weighted average (perpetual)

- COGS = **₹341,666.67**
- Ending inventory = **₹135,333.33**
- Gross profit = **₹138,333.33**

EXECUTIVE SUMMARY

GADGET HUB

JUNE 2025

FINANCIAL PERFORMANCE (JUNE 2025)

- Initial capital injection: ₹1,000,000
- Sales Revenue: ₹150,000
- Cost of Goods Sold: ₹90,000
- Marketing Expenses: ₹20,000

NET RESULTS (JUNE 2025)

- Total Assets ₹1,040,000
- Total Liabilities: ₹140,000
- Net Income: ₹40,000

INTERPRETATION

Gadget Hub generated a modest profit in its first month of operation. The company should focus on reducing costs and increasing sales to improve profitability and strength its financial position.

QUESTION 3 ~APEX SOLUTIONS INC.~ — FINANCIAL ANALYSIS (2022–2024)

1 HORIZONTAL ANALYSIS

(percentage change from prior year)

Income statement — 2023 vs 2022, and 2024 vs 2023

- Revenue: 2023 ↑ 20.00%; 2024 ↑ 25.00%
- Cost of Services: 2023 ↑ 11.11%; 2024 ↑ 20.00%
- Gross Profit: 2023 ↑ 27.27%; 2024 ↑ 28.57%
- Operating Expenses: 2023 ↑ 16.67%; 2024 ↑ 28.57%
- Interest Expense: 2023 ↑ 33.33%; 2024 ↑ 25.00%
- Net Income Before Tax: 2023 ↑ 40.91%; 2024 ↑ 29.03%
- Tax Expense: 2023 ↑ 40.91%; 2024 ↑ 29.03%
- Net Income After Tax: 2023 ↑ 40.91%; 2024 ↑ 29.03%

Brief note: revenue and net income grew strongly each year; net income growth outpaced revenue growth in 2023 and continued healthy growth in 2024.

- **Balance sheet — 2023 vs 2022, and 2024 vs 2023**
- Cash: 2023 ↑ 33.33%; 2024 ↑ 25.00%
- Accounts Receivable: 2023 ↑ 25.00%; 2024 ↑ 20.00%
- Other Current Assets: 2023 ↑ 33.33%; 2024 ↑ 25.00%
- Total Current Assets: 2023 ↑ 28.95%; 2024 ↑ 22.45%
- Property, Plant & Equipment (net): 2023 ↑ 25.00%; 2024 ↑ 20.00%
- Total Assets: 2023 ↑ 26.27%; 2024 ↑ 20.81%
- Accounts Payable: 2023 ↑ 33.33%; 2024 ↑ 25.00%
- Short-term Debt: 2023 ↑ 20.00%; 2024 ↑ 25.00%

- Total Current Liabilities: 2023 ↑ 25.00%; 2024 ↑ 25.00%
- Long-term Debt: 2023 ↑ 33.33%; 2024 ↑ 37.50%
- Shareholder's Equity: 2023 ↑ 23.61%; 2024 ↑ 12.36%

Brief note: assets and liabilities increased significantly each year. Long-term debt rose faster in 2024 than equity, indicating more debt-funded growth in 2024.

2 VERTICAL ANALYSIS

(Each line as a % of revenue for income statement; as a % of total assets for balance sheet)

Income statement (percent of Revenue)

2024 (base ₹15,000,000)

- Revenue = 100.00%
- Cost of Services = 40.00%
- Gross Profit = 60.00%
- Operating Expenses = 30.00%
- Interest Expense = 3.33%
- Net Income Before Tax = 26.67%
- Tax Expense = 8.00%
- Net Income After Tax = 18.67%

2023 (base ₹12,000,000)

- Cost of Services = 41.67%; Gross Profit = 58.33%; Operating Expenses = 29.17%; Interest = 3.33%; Net Income After Tax = 18.08%

2022 (base ₹10,000,000)

- Cost of Services = 45.00%; Gross Profit = 55.00%; Operating Expenses = 30.00%; Interest = 3.00%; Net Income After Tax = 15.40%

Trend takeaway (income): gross margin improved (55% → 58.33% → 60%) as cost of services declined as a share of revenue. Operating expenses stayed roughly stable as a percent of revenue (around 29–30%). Net margin improved from 15.40% (2022) to 18.67% (2024).

Balance sheet (percent of Total Assets)

2024 (Total assets ₹18,000,000)

- Cash = 13.89%
- Accounts Receivable = 16.67%
- Other Current Assets = 2.78%
- Total Current Assets = 33.33%
- PP&E (net) = 66.67%
- Accounts Payable = 5.56%
- Short-term Debt = 8.33%
- Total Current Liabilities = 13.89%
- Long-term Debt = 30.56%
- Shareholder's Equity = 55.56%

2023 (Total assets ₹14,900,000)

- Total Current Assets = 32.89%; PP&E = 67.11%; Total Current Liabilities = 13.42%; Long-term Debt = 26.85%; Equity = 59.73%

2022 (Total assets ₹11,800,000)

- Total Current Assets = 32.20%; PP&E = 67.80%; Total Current Liabilities = 13.56%; Long-term Debt = 25.42%; Equity = 61.02%

Trend takeaway (balance sheet): the asset mix is consistently capital-intensive (roughly two-thirds PP&E). Current assets are about one-third. Equity as a percentage of assets declined (61.02% → 59.73% → 55.56%) while long-term debt increased (25.42% → 26.85% → 30.56%) — company is using more long-term debt relative to equity to finance growth.

3 RATIO ANALYSIS — 2024 (CALCULATED)

Given figures / assumptions: Average Accounts Receivable = ₹2,750,000; Outstanding shares = 500,000.

- Net Profit Margin = Net Income After Tax / Revenue
  = 2,800,000 / 15,000,000 = 18.67%

- Quick Ratio = (Cash + Accounts Receivable + Other Current Assets) / Total Current Liabilities
  = (2,500,000 + 3,000,000 + 500,000) / 2,500,000 = 6,000,000 / 2,500,000 = 2.40
- Debt to Equity Ratio = (Short-term Debt + Long-term Debt) / Shareholder's Equity
  = (1,500,000 + 5,500,000) / 10,000,000 = 7,000,000 / 10,000,000 = 0.70
- Accounts Receivable Turnover = Revenue / Average Accounts Receivable
  = 15,000,000 / 2,750,000 = 5.45 times
- Earnings Per Share (EPS) = Net Income After Tax / Outstanding Shares
  = 2,800,000 / 500,000 = ₹5.60

4 TREND ANALYSIS (COST STRUCTURE AND FINANCIAL POSITION)

- **Cost structure:** Cost of Services fell as a percentage of revenue from 45.00% (2022) to 40.00% (2024). That raised gross margin from 55.00% to 60.00% over three years. Operating expenses remained around 29–30% of revenue, so the primary margin improvement comes from lower relative cost of services (better scale, pricing, or lower direct costs).
- **Profitability:** Net margin improved from 15.40% (2022) to 18.67% (2024). Net income grew faster than revenue in 2022→2023 and continued solid growth in 2024.
- **Liquidity:** Current assets consistently around one-third of total assets; quick ratio in 2024 is strong at 2.40, indicating more than adequate short-term liquidity.
- **Solvency / leverage:** Long-term debt rose as a share of assets; debt-to-equity is 0.70 in 2024 (higher absolute debt but still moderate). Equity as a share of assets fell from ~61% to ~55.6% — leverage has increased but remains manageable.
- **Working capital efficiency:** Accounts' receivable turnover at 5.45x lags the industry average (6.5x). That implies slower collection or more credit sales relative to peers.

5 INTERPRETATIONS, COMPARISON TO INDUSTRY AVERAGES, AND CONCLUSION

Comparison to industry (2024 averages you supplied):

- Industry Net Profit Margin = 22% ; Apex = 18.67% → below industry by ~3.33 percentage points. Apex is profitable but behind top industry margin.

- Industry Quick Ratio = 1.2 : 1 ; Apex = 2.40 → significantly stronger liquidity than industry.
- Industry Debt to Equity = 0.7 : 1 ; Apex = 0.70 → matches industry average.
- Industry Accounts Receivable Turnover = 6.5x ; Apex = 5.45x → slower collections than peers.
- Industry EPS = ₹6.00 ; Apex EPS = ₹5.60 → slightly below industry EPS.

Overall interpretation:

- **Strengths**
    - Strong revenue growth (20% → 25% YoY growth rates), and net income growth accelerating in prior years.
    - Improving gross margin (60% in 2024) driven by a lower proportion of cost of services.
    - Very healthy liquidity position (quick ratio 2.40), which reduces short-term risk.
    - Leverage (debt-to-equity 0.70) is in line with industry norms; financing appears deliberately debt-funded to support growth.
- **Areas for improvement / risks**
    - Net profit margin and EPS are below industry averages; operating efficiencies or pricing could be reviewed to close the gap.
    - Accounts receivable turnover is weaker than industry; this suggests slower collections or more liberal credit terms. That ties up cash and may reduce returns on assets.
    - Long-term debt increased materially in 2024 (long-term debt rose 37.5% vs prior year), which increases fixed financial obligations; monitor interest coverage and future cash flow to service debt.
- **Actionable recommendations**

1. Tighten receivables management: shorten collection terms, strengthen follow-up, consider incentives for early payment — this should raise AR turnover and free cash.
2. Examine billing and pricing strategy to push net margin toward industry levels (22%). Options: raise service prices where demand allows; identify further direct cost savings.
3. Monitor the cost and use of long-term debt. Since long-term debt rose in 2024, ensure returns on PP&E-funded projects exceed the after-tax cost of debt.

4. Maintain the strong liquidity cushion but use excess cash strategically (capex with good returns, or targeted buyback/dividend policy once margins stabilize).

Conclusion:

Apex Solutions shows solid, consistent growth with improving gross margins and strong liquidity. Profitability improved but remains slightly below industry leaders. The company has taken on more long-term debt to finance expansion; leverage is within industry norms but should be monitored. If management tightens receivables and pursues modest margin improvements, Apex is well positioned to close the performance gap with peers and sustain growth.

EXECUTIVE SUMMARY

FINANCIAL PERFORMANCE

- Revenue and net income have grown consistently
- Gross margin improved from 55% to 60% over three years

KEY RATIOS - 2024

Net Profit Margin	**18.67%**
Quick Ratio	**2.40**
Debt to Equity Ratio	**0.70**
Accounts Receivable Turnover	5.45

REVENUE, GROSS PROFIT, AND NET INCOME

INTERPRETATION AND CONCLUSION

Apex Solutions shows solid growth and improving profitability, but net margin and AR turnover lag industry averages. Recommendations; tighten collections and improve operational efficiency.